

Optical Fibre Losses

Attenuation in Optical Fibres

- Attenuation limits the optical power which can reach the receiver, limiting the operating span of a system.
- Once the power of an optical pulse is reduced to a point where the receiver is unable to detect the pulse, an error occurs.
- **Attenuation** is mainly a result of:
- **Light Absorption**
- **Scattering of light**
- **Bending losses**
- **Attenuation** is defined as the ratio of optical input power (P_i) to the optical output power (P_o).
- The following equation defines signal attenuation as a unit of length :

$$\text{Attenuation } \alpha(dB / km) = -\frac{10}{L} \text{Log}_{10} \left(\frac{P_{out}}{P_{in}} \right)$$

➤ Types of Attenuation

Absorption Loss:

- Caused by the fiber itself or by impurities in the fiber, such as water and metals.

Scattering Loss:

- Intrinsic loss mechanism caused by the interaction of photons with the glass itself.

Bending loss:

- Loss induced by physical stress on the fiber.

1. Material Absorption Losses

- Material absorption is caused by absorption of photons within the fiber.
- – When a material is illuminated, photons can make the valence electrons of an atom transition to higher energy levels
- – Photon is destroyed, and the radiant energy is transformed into electric potential energy. This energy can then
 - • Be re-emitted (scattering)
 - • Frees the electron (photoelectric effects) (not in fibers)
 - • Dissipated to the rest of the material (transformed into heat)
- In an optical fiber Material Absorption is the optical power that is effectively converted to heat dissipation within the fiber.
- • Two types of absorption exist:
 - – **Intrinsic Absorption**, caused by interaction with one or more of the components of the glass.
 - – **Extrinsic Absorption**, caused by impurities within the glass.

- **Intrinsic Absorption** is caused by basic fiber material properties. If an optical fiber is absolutely pure, with no imperfections or impurities, then all absorption will be intrinsic. Intrinsic absorption in the ultraviolet region is caused by electronic absorption bands. **Intrinsic Absorption** occurs when a light particle (photon) interacts with an electron and excites it to a higher energy level.

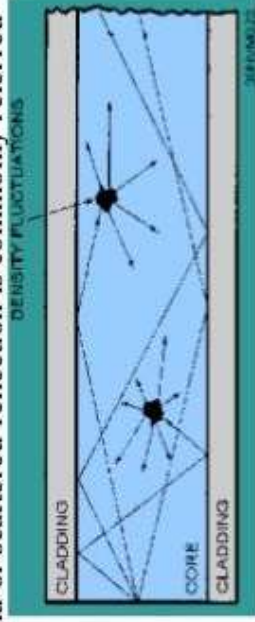
- **Extrinsic Absorption** is caused by impurities caused by impurities introduced into the fiber material. The metal impurities such as iron, nickel and chromium are introduced into the fiber during fabrication. **Extrinsic Absorption** is caused by the **electronic transition** of these metal ions from one energy level to another energy level.

2. Scattering Losses

- **Light scattering** is a form of scattering in which light in the form of propagating energy is scattered.
- Light scattering can be thought of as the deflection of a ray from a straight path, for example by irregularities in the propagation medium, particles, or in the interface between two media.
- Deviations from the law of reflection due to irregularities on a surface are also usually considered to be a form of scattering.
- When these irregularities are considered to be random and dense enough that their individual effects average out, this kind of scattered reflection is commonly referred to as diffuse reflection.

Linear Scattering may be of two types

- **Rayleigh Scattering**
- **Mie Scattering**

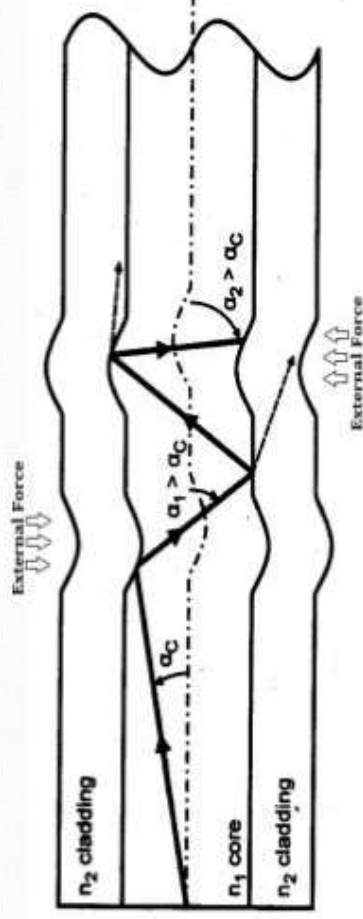


3. Fibre Bending losses

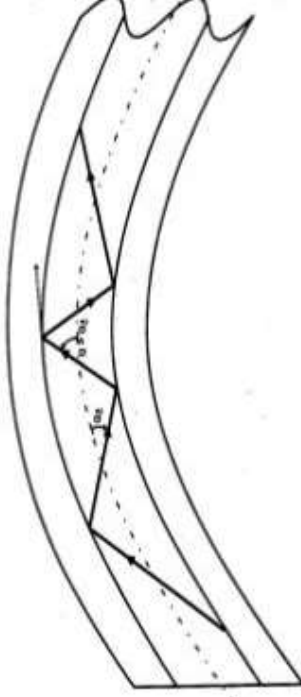
Bending loss is classified according to the bend radius of curvature :

1. **Microbend Loss**
2. **Macroband Loss**

➤ **Microbend Loss** are caused by small discontinuities or imperfections in the fiber. Uneven coating applications and improper cabling procedure increases micro bend loss. External forces are also a source of micro bends.

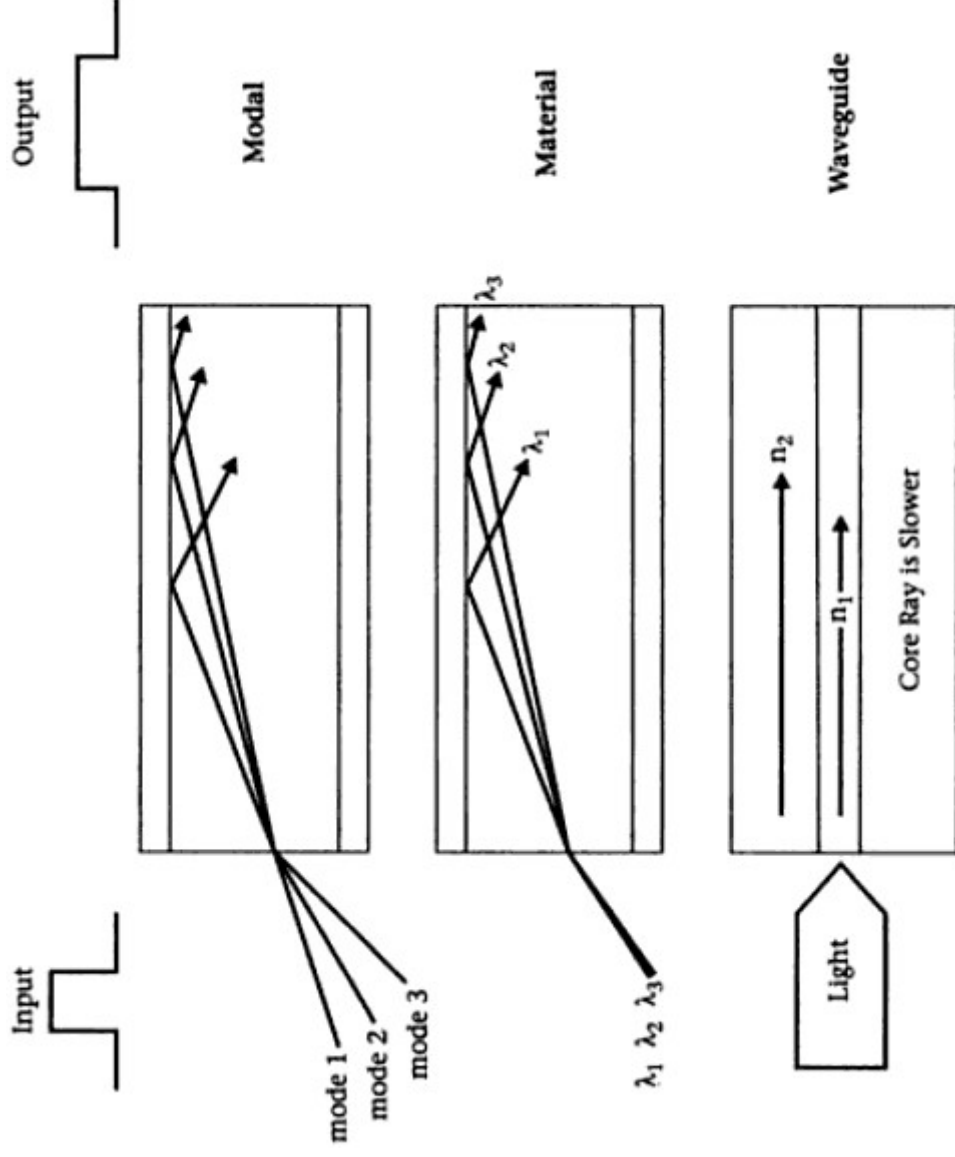


➤ **Macroband Losses** are observed when a fiber bend's radius of curvature is large compared to the fiber diameter. These bends are a great source of loss when the radius of curvature is less than several centimeters.



OPTICAL FIBER DISPERSION

Dispersion is the spreading out of a light pulse in time as it propagates down the fiber. Dispersion in optical fiber includes modal dispersion, material dispersion and waveguide dispersion. Each type is discussed in detail below.



MODEL DISPERSION IN MULTIMODE FIBERS

Inter model Dispersion

Multimode fibers can guide many different light modes since they have much larger core size. This is shown as the 1st illustration in the picture above. Each mode enters the fiber at a different angle and thus travels at different paths in the fiber.

Since each mode ray travels a different distance as it propagates, the rays arrive at different times at the fiber output. So the light pulse spreads out in time which can cause signal overlapping so seriously that you cannot distinguish them any more.

Model dispersion is not a problem in single mode fibers since there is only one mode that can travel in the fiber.

MATERIAL DISPERSION (Intra Model Dispersion)

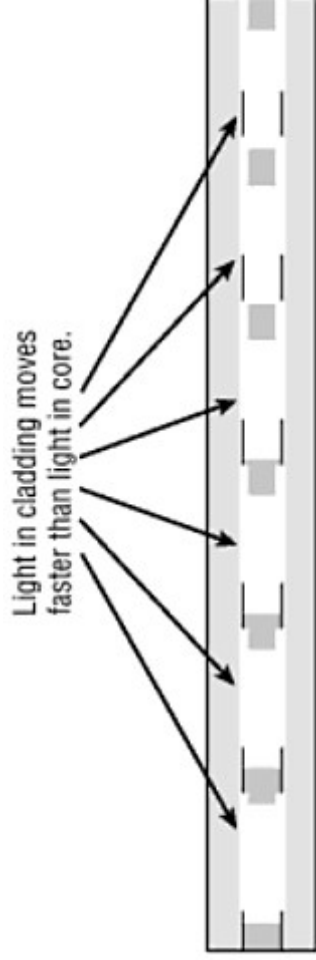
Material dispersion is the result of the finite linewidth of the light source and the dependence of refractive index of the material on wavelength. It is shown as the 2nd illustration in the first picture.

Material dispersion is a type of chromatic dispersion. Chromatic dispersion is the pulse spreading that arises because the velocity of light through a fiber depends on its wavelength.

WAVEGUIDE DISPERSION

Waveguide dispersion is only important in single mode fibers. It is caused by the fact that some light travels in the fiber cladding compared to most light travels in the fiber core. It is shown as the 3rd illustration in the first picture.

Since fiber cladding has lower refractive index than fiber core, light ray that travels in the cladding travels faster than that in the core. Waveguide dispersion is also a type of chromatic dispersion. It is a function of fiber core size, V-number, wavelength and light source linewidth.



While the difference in refractive indices of single mode fiber core and cladding are minuscule, they can still become a factor over greater distances.

Chromatic dispersion is a phenomenon that is an important factor in fiber optic communications. It is the result of the different colors, or wavelengths, in a light beam arriving at their destination at slightly different times

APPLICATIONS OF FIBER OPTICS

As the popularity of optical fibers continue to grow, so does their applications and practical uses. Fiber optic cables became more and more popular in a variety of industries and applications.

Communications / Data Storage

Since fiber optics are resistant to electronic noise, fiber optics has made significant advances in the field of communications. The use of light as its source of data transmission has improved the sound quality in voice communications. It is also being used for transmitting and receiving purposes.

Military

Optical systems offer more security than traditional metal-based systems. The magnetic interference allows the leak of information in the coaxial cables. Fiber optics is not sensitive to electrical interference; therefore fiber optics is suitable for military application and communications, where signal quality and security of data transmission are important.

The increased interest of the military in this technology caused the development of stronger fibers, tactical cables and high quality components. It was also applied in more varied areas such as hydrophones for seismic and SONAR, aircrafts, submarines and other underwater applications.

Medical

Fiber optic are used as light guides, imaging tools and as lasers for surgeries. Another popular use of fiber-optic cable is in an endoscope, which is a diagnostic instrument that enables users to see through small holes in the body. Medical endoscopes are used for minimally invasive exploratory or surgical procedures. Fiber optics is also used in bronchoscopes and laparoscopes.

All versions of endoscopes look like a long thin tube, with a lens or camera at one end through which light is emitted from the bundle of optical fibers banded together inside the enclosure.

Mechanical or Industrial

Industrial endoscopes also called a borescope or fiberscope, enables the user to observe areas that are difficult to reach or see under normal circumstances, such as jet engine interiors, inspecting mechanical welds in pipes and engines, inspecting space shuttles and rockets. Inspection of sewer lines and pipes.

Networking

Fiber optic is used to connect servers and users in a variety of network settings. It increases the speed, quality and accuracy of data transmission. Computer and Internet technology has improved due to the enhanced transmission of digital signals through optical fibers.

Industrial/Commercial

Fiber optics are used for imaging in areas which are difficult to reach. It is also used in wiring where electromagnetic interference is an problem. It gets used often as sensory devices to make temperature, pressure and other measurements as well as in the wiring of motorcars and in industrial settings.

Spectroscopy

Optical fiber bundles are used to transmit light from a spectrometer to a substance which cannot be placed inside the spectrometer itself, in order to analyse its composition. A spectrometer analyses substances by bouncing light off of and through them. By using optical fibers, a spectrometer can be used to study objects that are too large to fit inside, or gasses, or reactions which occur in pressure vessels

Broadcast/CATV/Cable Television

Broadcast or cable companies use fiber optic cables for wiring CATV, HDTV, internet, video and other applications.

Usage of fiber-optic cables in the cable-television industry began in 1976 and quickly spread because of the superiority of fiber optic cable over traditional coaxial cable. Fiber optic systems became less expensive and capable of transmitting clearer signals further away from the source signal. It also reduced signal losses and decreased the number of amplifiers required for each

customer. Fiber optic cable allows cable providers to offer better service, because only one optical line is needed for every ± 500 households.

Lighting and Imaging

Fiber optic cables are used for lighting and imaging and as sensors to measure and monitor a vast range of variables. It is also used in research, development and testing in the medical, technological and industrial fields.

Fiber optics are used as light guides in medical and other applications where bright light needs to shine on a target without a clear line-of-sight path. In some buildings, optical fibers are used to route sunlight from the roof to other parts of the building. Optical fiber illumination is also used for decorative applications, including signs, art and artificial Christmas trees.

Conclusion

With the introduction of highly transparent fiber-optic cable in the 1970s, very high-frequency laser signals now carry phenomenal loads of telephone conversations and data across the country and around the world.

From surgical procedures to worldwide communication via the internet, fiber optic has revolutionised our world. Fiber optics has made important contributions to the medical field, especially with regards to surgery. One of the most useful characteristics of optical fibers is their ability to enter the minute passageways and hard-to-reach areas of the human body. But perhaps the greatest contribution of the 20th century is the combination of fiber optics and electronics to transformed telecommunications.